

NICHOLAS BORRELLO

AI & Software Engineer — Agentic Systems & LLM Applications

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SKILLS & CAPABILITIES

AI & AGENTS

LLM agents • Model Context Protocol (MCP) • Tool calling • Structured outputs • Agent orchestration • Model routing/fallback • RAG • LLM evaluation • VLM/OCR

ML, NLP & LANGUAGES

TypeScript • Python • SQL • Java • PyTorch • CNNs • Transformers • spaCy • MediaPipe • Computer vision • Document intelligence

FULL STACK & INFRASTRUCTURE

React • Next.js • Node.js • Bun • PostgreSQL • SQLite • Supabase • Docker • Playwright • GitHub Actions • Azure DevOps

EXPERIENCE

Bay State Pet & Garden — AI & Software Engineer Taunton, MA (Remote) | Jul 2024 – Present

- ◆ Develop an AI Product Intelligence system that researches external product data, resolves exact product/variant identity, and produces evidence-backed catalog changes for human review.
- ◆ Engineer the agent execution layer with 25 bounded research tools, model routing and fallback, privacy controls, tool/model budgets, deadlines, and fail-closed validation.
- ◆ Maintain the retail CMS and AI evaluation pipeline, testing new extraction and agent behavior against verified datasets before staged rollout into product onboarding and publishing workflows.

Allegro MicroSystems — Process Engineering Intern Manchester, NH | Jun 2023 – Aug 2023

- ◆ Built an Azure DevOps CI/CD pipeline that automated driver deployment to engineering test equipment, replacing a manual deployment process.
- ◆ Developed supporting automation workflows to standardize driver updates across test systems and simplify repeatable engineering test runs.

EDUCATION

Worcester Polytechnic Institute — M.S. Computer Science Worcester, MA | Aug 2025

Worcester Polytechnic Institute — B.S. Computer Science, *With Distinction* Worcester, MA | May 2024

SELECTED PROJECTS

Resumancer AI — Agentic Career Workspace Oct 2025 – Present | Next.js, TypeScript, Vercel AI SDK

- ◆ Built a career workspace comparing job requirements with a Master CV, proposing evidence-backed changes with reviewable diffs instead of unconstrained rewrites.
- ◆ Added ATS compatibility checks, model fallback, and PDF export so verified career data moves from job analysis to an application-ready resume.

ShopSite MCP Server Jan 2026 | TypeScript, Model Context Protocol, Zod

- ◆ Built an MCP server that lets AI agents retrieve orders, search products, and manage inventory in ShopSite without interacting directly with legacy APIs.
- ◆ Wrapped XML/CGI requests, HMAC-SHA1 authentication, and legacy response formats behind typed, validated tools designed for safe LLM use.

Medusa Storefront Agent — Commerce Operations Jul 2026 | TypeScript, Medusa.js, PostgreSQL, MCP

- ◆ Built an autonomous store operations agent on Medusa.js that monitors inventory levels, drafts automated reorders, and applies pricing rules via typed MCP tool calls.
- ◆ Implemented Zod schema validation and PostgreSQL event listeners to coordinate reliable multi-step agent actions across inventory and pricing workflows.

American Sign Language Gesture Recognition Jan 2025 – May 2025 | Python, PyTorch, ResNet34, MediaPipe

- ◆ Trained CNN/Transformer models to recognize 100+ ASL gestures using ResNet34 and MediaPipe-derived features, and parallelized preprocessing for faster experimentation.
- ◆ Engineered real-time landmark extraction and temporal sequence processing pipelines, achieving robust classification across varying angles and lighting.